Samantha Lune Magid

Applied and Computational Mathematician

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Research Interests

Complex and adaptive systems; stochastic processes and history-dependent dynamics; science of science and metaresearch; scientific productivity and career trajectories; discrete-time count and sequential processes; emergent behavior in social and ecological systems; geometric and latent representations; causal inference and quasi-experimental identification; interpretable generative modeling; estuarine and intertidal systems; coupled air-sea climate systems; ethical limits of mathematical modeling.

Education

University of Vermont **Expected Spring 2028**

M.S. Complex Systems and Data Science, Vermont Complex Systems Institute *Burlington, VT*

- UVM Graduate Merit Scholar.
- Research areas: stochastic scientific productivity, science of science, metaresearch, and computational social science.

Boston University **B.S.**

B.S. Data Science, Applied and Mathematical Focus; Minor in Environmental Science *Boston, MA*

- Lara Vincent Undergraduate Research Scholar and Award recipient, October 2022.

Research Experience

Graduate Researcher **May 2026–Present**

Science and Humanity Lab, Vermont Complex Systems Institute *Burlington, VT*

- Develop scholar-agnostic stochastic models of scientific productivity to test whether persistent career differences can emerge from dynamic processes rather than fixed individual traits.
- Fit and simulate autoregressive, hurdle, self-exciting, and history-dependent models using longitudinal publication data from DBLP and the Academic Analytics Research Center.
- Identified distinct extensive and intensive margins of productivity, geometric memory decay, declining restart hazards after inactivity, and persistent but dynamically generated rank structure.
- Design distributional, predictive, and trajectory-level evaluations using held-out simulation, bootstrap confidence intervals, cross-validation, rank-mixing analyses, goodness-of-fit tests, and adversarial null models.
- Built reproducible Python pipelines for data preparation, model fitting, simulation, diagnostics, and publication-quality visualization across multiple linked research repositories.
- Contribute to manuscript development, figures, technical reports, conference materials, and a multi-field extension of prior work on scientific productivity as a stochastic process.

Manuscripts and Presentations

Zhang, S.; Magid, S.; Kumar, A.; Larremore, D. B.; Clauset, A. "Stochastic dynamics can drive large differences in researcher productivity." *Manuscript in preparation.*

Related findings were presented by Sam Zhang as a lightning talk at the *5th International Conference on the Science of Science and Innovation (ICSSI 2026)*, Boulder, Colorado, June 29, 2026.

The Productivity Garden. Independent paper in progress developing a stochastic project-pipeline model of publication timing, attrition, memory, and clustered output.

Research Software

Selected repository: spaarw_followup — model lineage, memory tests, trajectory simulation, empirical diagnostics, and reproducible figures for the scientific-productivity project. Additional public code includes seven_models_shockley and AARC-DBLP-comparison.

Fellowships and Outreach

VALI AI Fellow **2026–Present**

Vermont AI Literacy Initiative, Vermont Complex Systems Institute *Burlington, VT*

- Contribute to an interdisciplinary AI literacy initiative developing accessible, hands-on curricula for Vermont schools and communities.
- Support critical understanding of how AI systems work, their limitations and biases, and their broader social implications.

Laboratory and Teaching Experience

Graduate Laboratory Supervisor / Advanced Lab Professional II **July 2026–Present**

Human Cell Biology Program, University of Vermont *Burlington, VT*

- Run the hands-on operations of the Human Cell Biology preparation laboratory, coordinating materials, protocols, equipment, and readiness for instructional labs.
- Supervise preparation-lab technicians and coordinate with faculty and staff to support safe, consistent, and technically reliable laboratory instruction.

Undergraduate Research

Undergraduate Principal Investigator **September 2022–March 2023**

Uca spp. Bioturbation Survey, Boston University Marine Program *Boston, MA*

- Proposed and led an independent study of fiddler-crab overpopulation as a complex biophysical system, examining how stochastic bioturbation feedback cycles contribute to salt-marsh erosion.
- Designed and fabricated a custom tidal-forcing generator for controlled simulation of periodic intertidal conditions without disrupting field sites.
- Coordinated with marine scientists and field experts, completed the project's first phase, and produced reusable tools for future estuarine-modeling research.

Mathematical and Computational Skills

Stochastic and complex systems: Stochastic and Markov processes; discrete-time count and sequential models; autoregressive, hurdle, survival, hazard, and history-dependent models; geometric memory kernels; feedback-aware dynamical systems; emergence and nonlinearity; stochastic biological systems; latent state-space models.

Statistical and computational methods: Monte Carlo simulation; bootstrap inference; held-out simulation; cross-validation; goodness-of-fit testing; time-series analysis; low-dimensional approximations; rank-persistence and transition analysis; robustness and sensitivity analysis.

Optimization, learning, and causal methods: Discrete and continuous optimization; kernel methods; geometric representation; neural-network architectures; black-box optimization; simulated annealing; causal inference; quasi-experimental identification; adversarial and game-theoretic models.

Programming: Python, R, SQL, Rust.

Python scientific stack: pandas, NumPy, SciPy, statsmodels, scikit-learn, JAX, NumPyro, PyTorch, TensorFlow, XGBoost, Matplotlib, seaborn.

Research and visualization tools: Git, GitHub, Jupyter, Quarto, LaTeX, QGIS, ArcGIS, NetCDF, MySQL, Power BI, Tableau.

Laboratory and field: Preparation-lab supervision; wet-lab techniques; animal handling; laboratory safety; IACUC certification; experimental-system design.

Affiliations

Member, American Statistical Association (AmStat) | Member, Society for Industrial and Applied Mathematics (SIAM) | Vermont Complex Systems Institute | Science and Humanity Lab